

Q1 2021 (01 Sep Shift 2)

If $y = y(x)$ is the solution curve of the differential equation $x^2 dy + \left(y - \frac{1}{x}\right) dx = 0$; $x > 0$ and $y(1) = 1$, then $y\left(\frac{1}{2}\right)$ is equal to :

(1) $\frac{3}{2} - \frac{1}{\sqrt{e}}$

(2) $3 + \frac{1}{\sqrt{e}}$

(3) $3 + e$

(4) $3 - e$

Q2 2021 (31 Aug Shift 2)

If $y \frac{dy}{dx} = x \left[\frac{y^2}{x^2} + \frac{\phi\left(\frac{y^2}{x^2}\right)}{\phi\left(\frac{y^2}{x^2}\right)} \right]$, $x > 0$, $\phi > 0$, and $y(1) = -1$,

then $\phi\left(\frac{y^2}{4}\right)$ is equal to :

(1) $4\phi(2)$

(2) $4\phi(1)$

(3) $2\phi(1)$

(4) $\phi(1)$

Q3 2021 (31 Aug Shift 2)

If $\frac{dy}{dx} = \frac{2^x y + 2^y \cdot 2^x}{2^x + 2^{x+y} \log_e 2}$, $y(0) = 0$, then for $y = 1$, the value of x lies in the interval:

(1) $(1, 2)$

(2) $\left(\frac{1}{2}, 1\right]$

(3) $(2, 3)$

(4) $\left(0, \frac{1}{2}\right]$

Q4 2021 (31 Aug Shift 1)

If $\frac{dy}{dx} = \frac{2^{x+y}-2^x}{2^y}$, $y(0) = 1$, then $y(1)$ is equal to :

(1) $\log_2(2 + e)$

(2) $\log_2(1 + e)$

(3) $\log_2(2e)$

(4) $\log_2(1 + e^2)$

Q5 2021 (27 Aug Shift 2)

If the solution curve of the differential equation $(2x - 10y^3) dy + y dx = 0$, passes through the points $(0, 1)$ and $(2, \beta)$, then β is a root of the equation:

(1) $y^5 - 2y - 2 = 0$

(2) $2y^5 - 2y - 1 = 0$

(3) $2y^5 - y^2 - 2 = 0$

(4) $y^5 - y^2 - 1 = 0$

Q6 2021 (27 Aug Shift 2)

A differential equation representing the family of parabolas with axis parallel to y-axis and whose length of latus rectum is the distance of the point $(2, -3)$ from the line $3x + 4y = 5$, is given by :

(1) $10 \frac{d^2y}{dx^2} = 11$

(2) $11 \frac{d^2x}{dy^2} = 10$

(3) $10 \frac{d^2x}{dy^2} = 11$

(4) $11 \frac{d^2y}{dx^2} = 10$

Q7 2021 (27 Aug Shift 1)

Let us consider a curve, $y = f(x)$ passing through the point $(-2, 2)$ and the slope of the tangent to the curve at any point $(x, f(x))$ is given by $f(x) + xf'(x) = x^2$ Then :

(1) $x^2 + 2xf(x) - 12 = 0$

(2) $x^3 + xf(x) + 12 = 0$

$$(3) x^3 - 3xf(x) - 4 = 0$$

$$(4) x^2 + 2xf(x) + 4 = 0$$

Q8 2021 (27 Aug Shift 1)

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 2(y + 2 \sin x - 5)x - 2 \cos x$ such that $y(0) = 7$. Then $y(\pi)$ is equal to :

(1) $2e^{\pi^2} + 5$

(2) $e^{\pi^2} + 5$

(3) $3e^{\pi^2} + 5$

(4) $7e^{\pi^2} + 5$

Q9 2021 (26 Aug Shift 2)

Let $y(x)$ be the solution of the differential equation

$$2x^2 dy + (e^y - 2x) dx = 0, x > 0. \text{ If } y(e) = 1, \text{ then } y(1) \text{ is equal to :}$$

(1) 0

(2) 2

(3) $\log_e 2$

(4) $\log_e(2e)$

Q10 2021 (26 Aug Shift 1)

Let $y = y(x)$ be a solution curve of the differential equation $(y + 1) \tan^2 x dx + \tan x dy + y dx = 0$,

$$x \in \left(0, \frac{\pi}{2}\right).$$

If $\lim_{x \rightarrow 0^+} xy(x) = 1$, then the value of $y\left(\frac{\pi}{4}\right)$ is :

(1) $-\frac{\pi}{4}$

(2) $\frac{\pi}{4} - 1$

(3) $\frac{\pi}{4} + 1$

(4) $\frac{\pi}{4}$

Answer Key

Q1 (4)

Q2 (2)

Q3 (1)

Q4 (2)

Q5 (4)

Q6 (4)

Q7 (3)

Q8 (1)

Q9 (3)

Q10 (4)

Q1 (4)

$$x^2 dy + \left(y - \frac{1}{x}\right) dx = 0 : x > 0, y(1) = 1$$

$$x^2 dy + \frac{(xy-1)}{x} dx = 0$$

$$x^2 dy = \frac{(xy-1)}{x} dx$$

$$\frac{dy}{dx} = \frac{1-xy}{x^3}$$

$$\frac{dy}{dx} = \frac{1}{x^3} - \frac{y}{x^2}$$

$$\frac{dy}{dx} = \frac{1}{x^2} \cdot y = \frac{1}{x^3}$$

$$\text{If } e^{\int \frac{1}{x^2} dx} = e^{-\frac{1}{x}}$$

$$ye^{-\frac{1}{x}} = \int \frac{1}{x^3} \cdot e^{-\frac{1}{x}}$$

$$ye^{-\frac{1}{x}} = e^{-\frac{1}{x}} \left(1 + \frac{1}{x}\right) + C$$

$$1 \cdot e^{-1} = e^{-1}(2) + C$$

$$C = -e^{-1} = -\frac{1}{e}$$

$$ye^{-\frac{1}{x}} = e^{-\frac{1}{x}} \left(1 + \frac{1}{x}\right) - \frac{1}{e}$$

$$y\left(\frac{1}{2}\right) = 3 - \frac{1}{e} \times e^2$$

$$y\left(\frac{1}{2}\right) = 3 - e$$

Q2 (2)

Let, $y = tx$

$$\frac{dy}{dx} = t + x \frac{dt}{dx}$$

$$\therefore tx \left(t + x \frac{dt}{dx}\right) = x \left(t^2 + \frac{\varphi'(t^2)}{\varphi'(t^2)}\right)$$

$$t^2 + xt \frac{dt}{dx} = t^2 + \frac{\varphi'(t^2)}{\varphi'(t^2)}$$

$$\int \frac{t\varphi'(t^2)}{\varphi'(t^2)} dt = \int \frac{dx}{x}$$

$$\text{Let } \varphi(t^2) = p$$

$$\therefore \varphi'(t^2) 2t dt = dp$$

$$\Rightarrow \int \frac{dy}{2p} = \int \frac{dx}{x}$$

$$\frac{1}{2} \ln \varphi(t^2) = \ln x + \ln c$$

$$\varphi(t^2) = x^2 k$$

$$\varphi\left(\frac{y^2}{x^2}\right) = kx^2, \varphi(1) = k$$

$$\varphi\left(\frac{y^2}{4}\right) = 4\varphi(1)$$

Q3 (1)

$$\frac{dy}{dx} = \frac{2^x(y+2^y)}{2^x(1+2^y\ln 2)}$$

$$\Rightarrow \int \frac{(1+2^y)\ln 2}{(y+2^y)} dy = \int dx$$

$$\Rightarrow \ln|y+2^y| = x + c$$

$$x = 0; y = 0 \Rightarrow c = 0$$

$$\Rightarrow x = \ln|y+2^y|$$

$$\Rightarrow \text{at } y = 1, x = \ln 3$$

$$\therefore 3 \in (e, e^2) \Rightarrow x \in (1, 2)$$

Q4 (2)

$$\frac{dy}{dx} = \frac{2^x 2^y - 2^x}{2^y}$$

$$2^y \frac{dy}{dx} = 2^x (2^y - 1)$$

$$\int \frac{2^y}{2^y - 1} dy = \int 2^x dx$$

$$\frac{\ln(2^y - 1)}{\ln 2} = \frac{2^x}{\ln 2} + C$$

$$\Rightarrow \log_2(2^y - 1) = 2^x \log_2 e + C$$

$$\therefore y(0) = 1 \Rightarrow 0 = \log_2 e + C$$

$$C = -\log_2 e$$

$$\Rightarrow \log_2(2^y - 1) = (2^x - 1) \log_2 e$$

$$\text{put } x = 1, \log_2(2^y - 1) = \log_2 e$$

$$2^y = e + 1$$

$$y = \log_2(e + 1) \text{ Ans.}$$

Q5 (4)

$$(2x - 10y^3) dy + y dx = 0$$

$$\Rightarrow \frac{dx}{dy} + \left(\frac{2}{y}\right)x = 10y^2$$

$$I.F. = e^{\int \frac{2}{y} dy} = e^{2\ln(y)} = y^2$$

Solution of D.E. is

$$\therefore x \cdot y = \int (10y^2) y^2 \cdot dy$$

$$xy^2 = \frac{10y^5}{5} + C \Rightarrow xy^2 = 2y^5 + C$$

$$\text{It passes through } (0, 1) \rightarrow 0 = 2 + C \Rightarrow C = -2$$

$$\therefore \text{Curve is } xy^2 = 2y^5 - 2$$

Now, it passes through $(2, \beta)$

$$2\beta^2 = 2\beta^5 - 2 \Rightarrow \beta^5 - \beta^2 - 1 = 0$$

$$\therefore \beta \text{ is root of an equation } y^5 - y^2 - 1 = 0 \text{ Ans.}$$

Q6 (4)

$$\alpha.R = \frac{|3(2)+4(-3)-5|}{5} = \frac{11}{5}$$

$$(x-h)^2 = \frac{11}{5}(y-k)$$

differentiate w.r.t ' x ': -

$$2(x-h) = \frac{11}{5} \frac{dy}{dx}$$

again differentiate

$$2 = \frac{11}{5} \frac{d^2y}{dx^2}$$

$$\frac{11 d^2y}{dx^2} = 10.$$

Q7 (3)

$$y + \frac{xdy}{dx} = x^2 \text{ (given)}$$

$$\Rightarrow \frac{dy}{dx} + \frac{y}{x} = x$$

$$\text{If } = e^{\int \frac{1}{x} dx} = x$$

Solution of DE

$$\Rightarrow y \cdot x = \int x \cdot x dx$$

$$\Rightarrow xy = \frac{x^3}{3} + \frac{c}{3}$$

Passes through $(-2, 2)$, so

$$-12 = -8 + c \Rightarrow c = -4$$

$$\therefore 3xy = x^3 - 4$$

$$\text{ie. } 3x \cdot f(x) = x^3 - 4$$

Q8 (1)

$$\frac{dy}{dx} - 2xy = 2(2 \sin x - 5)x - 2 \cos x$$

$$IF = e^{-x^2}$$

$$\text{so, } y \cdot e^{-x^2} = \int e^{-x^2} (2x(2 \sin x - 5) - 2 \cos x) dx$$

$$\Rightarrow y \cdot e^{-x^2} = e^{-x^2} (5 - 2 \sin x) + c$$

$$\Rightarrow y = 5 - 2 \sin x + c \cdot e^{x^2}$$

$$\text{Given at } x = 0, y = 7$$

$$\Rightarrow 7 = 5 + c \Rightarrow c = 2$$

$$\text{So, } y = 5 - 2 \sin x + 2e^{x^2}$$

$$\text{Now at } x = \pi,$$

$$y = 5 + 2e^{\pi^2}$$

Q9 (3)

$$2x^2 dy + (e^y - 2x) dx = 0$$

$$\frac{dy}{dx} + \frac{e^y - 2x}{2x^2} = 0 \Rightarrow \frac{dy}{dx} + \frac{e^y}{2x^2} - \frac{1}{x} = 0$$

$$e^{-y} \frac{dy}{dx} - \frac{e^{-y}}{x} = -\frac{1}{2x^2} \Rightarrow \text{Put } e^{-y} = z$$

$$\frac{-dz}{dx} - \frac{z}{x} = -\frac{1}{2x^2} \Rightarrow xdz + zdx = \frac{dx}{2x}$$

$$d(xz) = \frac{dx}{2x} \Rightarrow xz = \frac{1}{2} \log_e x + c$$

$$xe^{-y} = \frac{1}{2} \log_e x + c, \text{ passes through } (e, 1)$$

$$\Rightarrow C = \frac{1}{2}$$

$$xe^{-y} = \frac{\log_e ex}{2}$$

$$e^{-y} = \frac{1}{2} \Rightarrow y = \log_e 2$$

Q10 (4)

$$(y+1) \tan^2 x dx + \tan x dy + y dx = 0$$

$$\text{or } \frac{dy}{dx} + \frac{\sec^2 x}{\tan x} \cdot y = -\tan x$$

$$IF = e^{\int \frac{\sec^2 x}{\tan x} dx} = e^{\ln \tan x} = \tan x$$

$$\therefore y \tan x = - \int \tan^2 x dx$$

$$\text{or } y \tan x = - \tan x + x + C$$

$$\text{or } \lim_{x \rightarrow 0} xy = -x + \frac{x^2}{\tan x} + \frac{Cx}{\tan x} = 1$$

$$\text{or } \lim_{x \rightarrow 0} xy = -x + \frac{x^2}{\tan x} + \frac{Cx}{\tan x} = 1$$

$$\text{or } C = 1$$

$$y(x) = \cot x + x \cot x - 1$$

$$y\left(\frac{\pi}{4}\right) = \frac{\pi}{4}$$